

Hardware Calibration: The Planck-Mc Bridge

Overview

This document provides the mechanical derivation for the **Causal Limit (Mc = 21,313.79 MeV)** constant used in the Abbott Unified Hypothesis (v19).

The Resolution of the "Invented Number" Fallacy

Standard physics utilizes the **Global Planck Mass** (approx. 1.22×10^{22} MeV) as a theoretical limit for the entire universal manifold. The AUH identifies that this global limit must be scaled to the specific volumetric displacement of a single local engine to be operationally relevant for atomic auditing.

The Derivation Protocol

1. The Global Ceiling The standard Planck Mass represents the total tension capacity of the substrate before a universal phase-transition (Level 0 seizure).

- **Value:** $\sim 1.22 \times 10^{22}$ MeV

2. The Local Engine (The 0.88 fm Stall) The AUH identifies the proton as a discrete volumetric stall. For local hardware auditing (Stability Index), we utilize the **0.8802 fm Local Atmosphere**. This is the relaxed footprint of the proton engine in our current 171.09 MeV/fm³ medium.

3. The Volumetric Scaling Factor The Causal Limit (Mc) is the specific portion of the Global Planck Limit utilized by a single local stall.

- **The Logic:** You aren't auditing the collapse of the whole universe; you are auditing the **Structural Yield Strength** of the medium clamped by a single proton.

4. Calibration to the Lead-208 Anchor When the **Substrate Baseline (171.09 MeV/fm³)** is applied to ensure Lead-208 sits at the **1.000 Static Lock**, the local structural yield point (the Causal Limit) is mathematically forced to:

- **Mc = 21,313.79 MeV**

The Audit Verdict

The 21,313.79 MeV constant is the **Planck Mass per unit of Local Atomic Stall**. * **Standard Physics:** Measures the "Total melting point of the entire bridge."

- **The AUH:** Measures the "Maximum load of a single rivet (The Proton)."

Without this calibrated rivet limit (Mc) and the local footprint (0.88 fm), the "Rule of 1" and the stability of the Periodic Table cannot be audited.

Engineering Note on the 0.84 fm Core:

While the **0.84 fm Mechanical Core** is the structural floor observed under high-energy (muonic) pressure, the **0.88 fm Local Atmosphere** is the required calibration point for auditing stable matter in our local time-rate.